Active Inference for Joint Port Selection and Pilot Allocation in Fluid Antenna Systems

Kian Fotovat[1] and Kamran Fotovat[2]

***Abstract*—Fluid antenna systems (FAS) obtain spatial degrees of freedom by activating a few radiating ports from a dense grid of candidates, but deciding which ports to activate presupposes channel knowledge at ports that are never measured. Prior port-selection work assumes full or near-full channel state information (CSI) and treats the pilot overhead as a fixed cost. We instead formulate port selection and pilot allocation *jointly* as active inference. A spatio-temporal generative model — sub-wavelength spatial correlation together with an order-p autoregressive temporal prior — maintains a Gaussian belief over all N port channels, and the transmitter chooses both the ports it serves and the subset of them it pilots by minimizing an expected free energy that trades achievable rate against information gain and port-switching cost. Selection is greedy and submodular, at $\mathcal{O}(NM)$ per slot. On a 441-port grid with a hybrid front end at $n_{\mathrm{RF}} = 2K$, the agent attains 91% of a full-CSI genie's sum rate while measuring only 2.3% of the candidate ports; pilots can then be withdrawn from ports that are still served, one fifth of them for 2% of the sum rate and two fifths for 8%. No training is required.**

***Index Terms*—Fluid antenna systems, port selection, pilot allocation, active inference, expected free energy, partial CSI, Kalman filtering.**

I. Introduction

Fluid antenna systems (FAS), also called movable- or reconfigurable-position antenna systems, replace fixed radiating elements with elements whose position can be adjusted among a dense set of candidate positions, or ports, within a compact aperture [1]. By moving to a favourable point of the spatial fading pattern, an FAS harvests diversity that a fixed array of the same size cannot, using only a handful of radio-frequency (RF) chains, since at any instant only the activated ports are connected to the front end. The pivotal control problem is port selection: deciding slot by slot which subset of the candidates to activate.

A port can be scored only if its channel is known, so conventional selection presupposes channel state information (CSI) at every candidate. That is expensive. The aperture is sampled at sub-wavelength spacing, so the ports are packed far more densely than a half-wavelength array and the number of coefficients to estimate grows rapidly with aperture. Worse, only activated ports can be observed at all. The bottleneck is therefore not the selection arithmetic but channel acquisition under a sensing budget: the system must decide which ports are worth measuring — and, we add, whether a port that is served needs to be measured at all.

Two lines of work approach this and leave the gap open. Learning-based selection — deep reinforcement learning over ports [2] and online bandits [3] — learns policies directly, but treats the full channel as an observed, cost-free input and requires extensive training. Channel estimation for FAS [4], [5] exploits spatial correlation to reconstruct the channel from few pilots, but solves estimation in isolation: it recovers the channel without deciding what to sense or what to serve, and never weighs a measurement against its cost. What is missing is a formulation in which sensing and serving are chosen jointly under an explicit budget.

Active inference supplies that perspective. Rooted in the free-energy principle [6], it casts a decision-maker as an agent holding a generative model of its environment and acting to minimize an expected free energy (EFE), which decomposes into a pragmatic value rewarding goal attainment and an epistemic value rewarding informative actions. The fit to FAS is close: the hidden state is the channel at unmeasured ports, the generative model is its spatial and temporal correlation, the pragmatic value is throughput, and the epistemic value is what a pilot reveals about the ports left unseen. The exploit–explore tension that defines the problem is thus expressed by the EFE rather than engineered in by hand.

1) We formulate port selection *and pilot allocation* jointly as active inference under partial CSI. The piloted set is a strict subset of the activated set, making the pilot budget a decision variable rather than a fixed cost — to our knowledge the first such treatment for FAS.
2) We give a concrete agent: a Gaussian belief over all N ports from a spatial prior and an order-p autoregressive temporal model, with ports and pilots chosen by greedy EFE minimization at $\mathcal{O}(NM)$ per slot, and an exact reduced-rank belief that makes a 21×21 grid tractable.
3) We show that the agent reaches 91% of a full-CSI genie's sum rate while measuring 2.3% of the ports, and degrades gracefully as pilots are further withdrawn from ports it continues to serve.

II. Fluid Antenna System Model

We consider the downlink of a single-cell multiuser system in which a base station (BS) equipped with a fluid antenna serves K single-antenna users. This section specifies the array geometry, the spatio-temporal channel, the partial observation model that is central to this work, the transmission and rate model, and the switching-aware objective.

[1]Dept. of Electrical and Computer Engineering, University of Tehran, Tehran, Iran.
[2]Iran University of Science and Technology, Tehran, Iran.

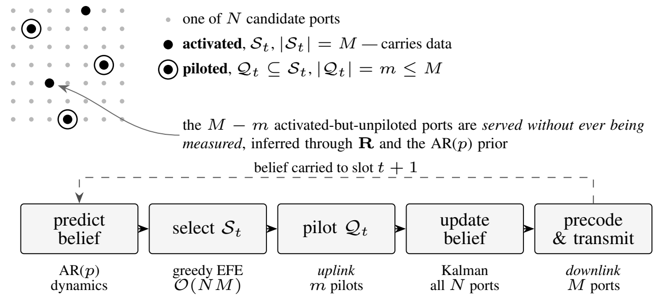

Fig. 1. Per-slot operation. The base station activates M of N candidate ports, but spends pilots on only $m \leq M$ of them; the remaining $M - m$ activated ports carry data while never being measured, their channels supplied by the belief alone. Both the activated set $\mathcal{S}_t$ and the piloted subset $\mathcal{Q}_t$ are chosen by minimizing the same expected free energy, which is what turns the pilot budget from a fixed cost into a design variable.

A. Array Geometry and Activation

The BS fluid antenna comprises $N = N_x \times N_y$ candidate ports arranged on a uniform planar grid over a compact aperture with sub-half-wavelength spacing. The BS has M radio-frequency (RF) chains, $M \leq N$, so in each slot t it activates a subset $\mathcal{S}(t) \subseteq \{1, \ldots, N\}$ of $|\mathcal{S}(t)| = M$ ports; only activated ports radiate and connect to the front end. Serving K streams requires $M \geq K$, hence $N \geq M \geq K$. We write $\mathbf{P}_\mathcal{S} \in \{0,1\}^{M \times N}$ for the selection matrix whose rows are the standard basis vectors indexed by $\mathcal{S}$, so that $\mathbf{P}_\mathcal{S}\mathbf{x}$ extracts the entries of a vector $\mathbf{x}$ on the activated ports.

B. Spatio-Temporal Channel Model

Let $\mathbf{h}_k(t) \in \mathbb{C}^N$ denote the channel between the N candidate ports and user k in slot t. Because the ports are densely spaced, their coefficients are spatially correlated. Following the classical Jakes model for rich scattering, the correlation between ports i and j depends only on their separation d_{ij},

$$R_{ij} = J_0\left(\frac{2\pi d_{ij}}{\lambda}\right), \tag{1}$$

where $J_0(\cdot)$ is the zeroth-order Bessel function of the first kind and λ the wavelength. Collecting these entries into $\mathbf{R} \in \mathbb{C}^{N \times N}$, the small-scale fading is spatially correlated Rayleigh, $\mathbf{h}_k \sim \mathcal{CN}(\mathbf{0}, \beta_k \mathbf{R})$, with β_k the large-scale gain of user k. The channel ages between slots according to a first-order Gauss–Markov (autoregressive) process,

$$\mathbf{h}_k(t) = \rho\, \mathbf{h}_k(t-1) + \sqrt{1-\rho^2}\, \mathbf{e}_k(t), \quad \mathbf{e}_k(t) \sim \mathcal{CN}(\mathbf{0}, \beta_k \mathbf{R}), \tag{2}$$

where $\mathbf{e}_k(t)$ is independent of $\mathbf{h}_k(t-1)$, and the temporal correlation $\rho = J_0(2\pi f_D T_s)$ is set by the maximum Doppler frequency f_D and the slot duration T_s; $\rho \to 1$ for slow fading and decreases with mobility. The process is stationary with marginal $\mathbf{h}_k(t) \sim \mathcal{CN}(\mathbf{0}, \beta_k \mathbf{R})$.

C. Observation Model: Partial CSI

A defining constraint of our setting is that the BS cannot observe the full channel: information about a port is obtained only when that port is activated. During the pilot phase of slot t the users send orthogonal pilots and the BS estimates the channel on the M activated ports, while the remaining $N - M$ ports are unobserved. The measurement is

$$\mathbf{y}_k(t) = \mathbf{P}_{\mathcal{S}(t)}\, \mathbf{h}_k(t) + \mathbf{n}_k(t), \quad \mathbf{n}_k(t) \sim \mathcal{CN}(\mathbf{0}, \sigma_e^2 \mathbf{I}_M), \tag{3}$$

where $\mathbf{P}_{\mathcal{S}(t)}$ restricts $\mathbf{h}_k(t)$ to the activated ports and $\mathbf{n}_k(t)$ is the pilot/estimation noise with per-port variance σ_e^2 set by the pilot SNR. Equivalently, the BS observes the length-M vector $\mathbf{h}_{k,\mathcal{S}}(t) = \mathbf{P}_{\mathcal{S}(t)}\mathbf{h}_k(t)$ in noise and must infer the $N - M$ hidden ports from spatial and temporal correlation. This partial observability distinguishes our formulation from prior port-selection work that assumes full-channel knowledge, and it is what makes the choice of which ports to measure consequential.

D. Downlink Transmission and Achievable Rate

Over the activated ports the BS transmits with a linear precoder $\mathbf{W}(t) = [\mathbf{w}_1(t), \ldots, \mathbf{w}_K(t)] \in \mathbb{C}^{M \times K}$, one column per user, under a total power budget $\|\mathbf{W}(t)\|_F^2 \leq P$. The transmitted signal is $\mathbf{x}(t) = \sum_k \mathbf{w}_k(t) s_k(t)$ with data symbols $s_k \sim \mathcal{CN}(0,1)$. User k receives

$$r_k = \mathbf{h}_{k,\mathcal{S}}^{\mathrm{H}} \mathbf{w}_k s_k + \sum_{j \neq k} \mathbf{h}_{k,\mathcal{S}}^{\mathrm{H}} \mathbf{w}_j s_j + z_k, \tag{4}$$

where $\mathbf{h}_{k,\mathcal{S}} = \mathbf{P}_\mathcal{S}\mathbf{h}_k$ is the channel on the activated ports and $z_k \sim \mathcal{CN}(0, \sigma^2)$ is receiver noise. Treating inter-user interference as noise, the SINR and achievable rate of user k are

$$\mathrm{SINR}_k = \frac{\left|\mathbf{h}_{k,\mathcal{S}}^{\mathrm{H}} \mathbf{w}_k\right|^2}{\sum_{j \neq k} \left|\mathbf{h}_{k,\mathcal{S}}^{\mathrm{H}} \mathbf{w}_j\right|^2 + \sigma^2}, \tag{5}$$

$$R_k = \log_2(1 + \mathrm{SINR}_k)\,. \tag{6}$$

The precoder is obtained by regularized MMSE (transmit-Wiener) filtering from the available channel estimate $\hat{\mathbf{H}} = [\hat{\mathbf{h}}_{1,\mathcal{S}}, \ldots, \hat{\mathbf{h}}_{K,\mathcal{S}}] \in \mathbb{C}^{M \times K}$,

$$\mathbf{W} = \xi\, \hat{\mathbf{H}} \left(\hat{\mathbf{H}}^{\mathrm{H}}\hat{\mathbf{H}} + \frac{K\sigma^2}{P}\mathbf{I}_K\right)^{-1}, \tag{7}$$

where ξ scales $\mathbf{W}$ to meet the power budget. Under full CSI, $\hat{\mathbf{H}}$ is the true activated channel; in our partial-CSI setting $\hat{\mathbf{H}}$ and its error statistics are supplied by the belief of Section III, yielding a robust MMSE precoder that additionally accounts for the residual channel uncertainty.

E. Switching Cost and Problem Formulation

Reconfiguring the fluid antenna between slots is not free: physically moving or re-tuning elements incurs delay and energy. We model this by the number of ports that change between consecutive slots,

$$C_t = \left|\mathcal{S}_t \triangle \mathcal{S}_{t-1}\right|, \qquad E_{\mathrm{sw}}(t) = e_{\mathrm{sw}}\, C_t, \tag{8}$$

where $\triangle$ is the symmetric set difference and e_{sw} the per-port switching energy. The design goal is to choose the activation sequence that maximizes long-term throughput net of switching cost,

$$\max_{\{\mathcal{S}_t\}} \sum_{t=1}^{T} \left(\sum_{k=1}^{K} R_k(t) - \eta_{\mathrm{sw}} E_{\mathrm{sw}}(t)\right) \quad \text{s.t. } |\mathcal{S}_t| = M, \tag{9}$$

where $\eta_{\rm sw} \geq 0$ trades throughput against reconfiguration cost. Equation (9) is the objective against which every method in this paper is evaluated. Its difficulty is that $R_k(t)$ depends on the channel over ports that, under partial CSI, are largely unobserved — the problem we address next through active inference.

III. Active Inference for Port Selection

We now cast port selection as active inference. The agent holds a generative model of the channel, maintains a Bayesian belief over all ports (perception), and each slot chooses the port set that minimizes an expected free energy (action), which trades expected rate against information gain and switching cost within one objective.

A. Generative Model

The latent state is the full channel $\{\mathbf{h}_k(t)\}$, of which only the activated ports are observed. The agent's generative model factorizes over users and time as

$$p\big(\{\mathbf{h}_k, \mathbf{y}_k\}\big) = \prod_k p\big(\mathbf{h}_k(0)\big) \prod_t p\big(\mathbf{h}_k(t) \,|\, \mathbf{h}_k(t-1)\big) \times\ p\big(\mathbf{y}_k(t) \,|\, \mathbf{h}_k(t), \mathcal{S}_t\big), \tag{10}$$

with three ingredients already fixed by the physics of Section II: the prior $p(\mathbf{h}_k(0)) = \mathcal{CN}(\mathbf{0}, \beta_k \mathbf{R})$; the transition $p(\mathbf{h}_k(t)|\mathbf{h}_k(t-1))$ given by the AR(1) dynamics (2); and the likelihood $p(\mathbf{y}_k(t)|\mathbf{h}_k(t), \mathcal{S}_t)$ given by the partial observation (3). Preferences complete the model. Unlike textbook active inference, in which preferred outcomes are encoded in a hand-tuned vector, here the preference is the communication utility itself — the achievable rate net of switching in (9). There is thus no arbitrary preference to set: the agent's goal is the system objective.

B. Perception: Complex Kalman Belief

Because the model is linear-Gaussian, the exact posterior over each user's channel is Gaussian, $q(\mathbf{h}_k) = \mathcal{CN}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$, maintained by a per-user complex Kalman filter with two steps per slot. The predict step propagates the belief through the AR(1) dynamics, encoding channel aging,

$$\boldsymbol{\mu}_k \leftarrow \rho\, \boldsymbol{\mu}_k, \qquad \boldsymbol{\Sigma}_k \leftarrow \rho^2 \boldsymbol{\Sigma}_k + (1-\rho^2)\beta_k \mathbf{R}. \tag{11}$$

When the ports in $\mathcal{S}_t$ are activated and $\mathbf{y}_k$ observed, the measurement update corrects the belief. With Kalman gain

$$\mathbf{K}_k = \boldsymbol{\Sigma}_k \mathbf{P}_{\mathcal{S}}^{\rm H} \left(\mathbf{P}_{\mathcal{S}} \boldsymbol{\Sigma}_k \mathbf{P}_{\mathcal{S}}^{\rm H} + \sigma_e^2 \mathbf{I}_M\right)^{-1}, \tag{12}$$

the posterior mean and covariance become

$$\boldsymbol{\mu}_k \leftarrow \boldsymbol{\mu}_k + \mathbf{K}_k\big(\mathbf{y}_k - \mathbf{P}_{\mathcal{S}} \boldsymbol{\mu}_k\big), \tag{13}$$

$$\boldsymbol{\Sigma}_k \leftarrow \big(\mathbf{I} - \mathbf{K}_k \mathbf{P}_{\mathcal{S}}\big) \boldsymbol{\Sigma}_k \big(\mathbf{I} - \mathbf{K}_k \mathbf{P}_{\mathcal{S}}\big)^{\rm H} + \sigma_e^2 \mathbf{K}_k \mathbf{K}_k^{\rm H}. \tag{14}$$

where the Joseph form (14) keeps $\boldsymbol{\Sigma}_k$ symmetric positive semidefinite. Only the $M \times M$ innovation covariance is inverted, and it is regularized by $\sigma_e^2 \mathbf{I}$; consequently the rank-deficient spatial prior $\mathbf{R}$ needs no artificial jitter. Crucially, observing a port also shrinks the uncertainty of its correlated neighbours through the off-diagonals of $\mathbf{R}$ and $\boldsymbol{\Sigma}_k$, so a few measurements inform many ports. The filter is calibrated: its predicted covariance matches the realized estimation error.

C. Action: Expected Free Energy

Each slot the agent scores a candidate port set $\mathcal{S}$ by its expected free energy, written here as a value to be minimized,

$$G(\mathcal{S}) = -\,\mathrm{Prag}(\mathcal{S}) - \kappa\, \mathrm{Epis}(\mathcal{S}) + \eta_{\rm sw} E_{\rm sw}(\mathcal{S}), \tag{15}$$

a pragmatic (goal) term, an epistemic (information) term weighted by an exploration weight $\kappa \geq 0$, and the switching cost of (8). The pragmatic value is the expected sum rate the belief predicts for $\mathcal{S}$ under the robust MMSE precoder; accounting for residual CSI error through the belief covariance gives the imperfect-CSI lower bound

$$\mathrm{Prag}(\mathcal{S}) = \sum_k \log_2\left(1 + \frac{\left|\boldsymbol{\mu}_{k,\mathcal{S}}^{\rm H} \mathbf{w}_k\right|^2}{\mathcal{I}_k}\right), \tag{16}$$

$$\mathcal{I}_k = \sum_{j \neq k} \left|\boldsymbol{\mu}_{k,\mathcal{S}}^{\rm H} \mathbf{w}_j\right|^2 + \sum_j \mathbf{w}_j^{\rm H} \boldsymbol{\Sigma}_{k,\mathcal{S}} \mathbf{w}_j + \sigma^2, \tag{17}$$

where $\boldsymbol{\mu}_{k,\mathcal{S}} = \mathbf{P}_{\mathcal{S}} \boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}_{k,\mathcal{S}} = \mathbf{P}_{\mathcal{S}} \boldsymbol{\Sigma}_k \mathbf{P}_{\mathcal{S}}^{\rm H}$, and the error term $\sum_j \mathbf{w}_j^{\rm H} \boldsymbol{\Sigma}_{k,\mathcal{S}} \mathbf{w}_j$ makes the agent conservative when uncertain; the robust MMSE precoder augments (7) with $\boldsymbol{\Sigma}_{k,\mathcal{S}}$. The epistemic value is the mutual information between the channel and the pilots that $\mathcal{S}$ would yield, i.e., the expected reduction in belief entropy,

$$\mathrm{Epis}(\mathcal{S}) = \sum_k \log_2 \det\left(\mathbf{I}_M + \frac{\mathbf{P}_{\mathcal{S}} \boldsymbol{\Sigma}_k \mathbf{P}_{\mathcal{S}}^{\rm H}}{\sigma_e^2}\right), \tag{18}$$

which is monotone and submodular in $\mathcal{S}$; through the correlation in $\boldsymbol{\Sigma}_k$, activating one port also gathers information about others. The two terms embody the exploit–explore tension natively: the pragmatic value favours ports that serve well now, the epistemic value favours ports that most reduce uncertainty for later, and κ sets the balance — no separate exploration heuristic is bolted on.

D. Observe-Then-Precode Protocol

The order of operations within a slot matters. We adopt an observe-then-precode protocol: (i) predict the belief; (ii) select $\mathcal{S}_t$ by minimizing G on the predicted belief; (iii) activate $\mathcal{S}_t$, obtain the pilots $\mathbf{y}_k$, and update the belief; and (iv) form the robust MMSE precoder from the updated belief and transmit. Because the served ports are re-measured before precoding, their posterior error is about σ_e^2 rather than the one-step aging floor $(1-\rho^2)\beta_k$ that a predict-then-act ordering incurs. As shown in Section IV, this makes the served rate nearly independent of Doppler.

E. Greedy Submodular Selection

Selecting the best M-of-N set exactly is combinatorial. We instead build $\mathcal{S}_t$ greedily: starting from the empty set, we repeatedly add the port whose inclusion most decreases G,

$$p^\star = \underset{p \notin \mathcal{S}}{\arg\max}\ \Big[G(\mathcal{S}) - G\big(\mathcal{S} \cup \{p\}\big)\Big], \tag{19}$$

Algorithm 1 Active-Inference Port Selection (slot t)

Require: predicted belief $\{\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}$, previous set $\mathcal{S}_{t-1}$, budget M, weight κ
Ensure: activated set $\mathcal{S}_t$ and precoder $\mathbf{W}$
1: predict belief: $\boldsymbol{\mu}_k \leftarrow \rho\boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}_k \leftarrow \rho^2\boldsymbol{\Sigma}_k + (1-\rho^2)\beta_k\mathbf{R}$ $\forall k$
2: $\mathcal{S} \leftarrow \emptyset$
3: **for** $m = 1$ **to** M **do**
4:     **for** each candidate port $p \notin \mathcal{S}$ **do**
5:         $\Delta(p) \leftarrow G(\mathcal{S}) - G(\mathcal{S} \cup \{p\})$ $\quad\triangleright$ via (15)–(18)
6:     **end for**
7:     $p^\star \leftarrow \arg\max_p \Delta(p)$; $\mathcal{S} \leftarrow \mathcal{S} \cup \{p^\star\}$
8: **end for**
9: $\mathcal{S}_t \leftarrow \mathcal{S}$
10: activate $\mathcal{S}_t$, observe pilots $\mathbf{y}_k$, update belief $\quad\triangleright$ via (12)–(14)
11: $\mathbf{W} \leftarrow$ robust MMSE precoder from updated $\{\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}$
12: transmit; incur switching cost $\eta_{\text{sw}} \cdot |\mathcal{S}_t \triangle \mathcal{S}_{t-1}|$

until $|\mathcal{S}| = M$. Since the epistemic term is monotone submodular and the switching term modular, greedy selection attains the standard $(1-1/e)$ near-optimality guarantee on the information-driven part, at $\mathcal{O}(NM)$ score evaluations per slot — about 20 ms in our setting versus seconds for exhaustive search, a roughly $500\times$ reduction. Algorithm 1 summarizes the complete per-slot agent, with the greedy loop (lines 3–8) embedded in the observe-then-precode ordering of Section III-D.

F. Online Learning of the Spatial Correlation

The agent assumes a Jakes $\mathbf{R}$, but real propagation may differ. When $\mathbf{R}$ is unknown or non-Jakes it can be learned online from the accumulated pilots: averaging the outer products of co-observed pilots and removing the noise floor gives

$$\hat{R}_{ij} = \frac{\sum_t \mathbf{1}\{i,j \in \mathcal{S}_t\}\, y_i(t)\, y_j^*(t)}{\sum_t \mathbf{1}\{i,j \in \mathcal{S}_t\}} - \sigma_e^2\, \delta_{ij}, \tag{20}$$

which is then normalized to unit diagonal (cancelling the per-user power β_k) and projected onto the positive-semidefinite cone. Substituting $\hat{\mathbf{R}}$ for $\mathbf{R}$ restores performance under model mismatch (Section IV); in the matched case the belief is already Bayes-optimal, so learning can only help — consistent with the method's zero-training nature.

IV. Numerical Results

[TODO: Not yet written in the working draft. Figures exist in figures/.]

V. Conclusion

[TODO: Not yet written in the working draft.]

References

[1] K.-K. Wong, K.-F. Tong, Y. Zhang, and W. Zhongbin, "Fluid antenna system: A new spatial diversity paradigm," *IEEE Trans. Wireless Commun.*, 2021.

[2] "Switching-cost-aware deep reinforcement learning for dynamic port selection in fluid antenna systems," *IEEE Commun. Lett.*, 2024.

[3] J. Zou, S. Sun, and C. Wang, "Online learning-induced port selection for fluid antenna in dynamic channel environment," *IEEE Wireless Commun. Lett.*, 2024.

[4] Z. Zhang, J. Zhu, L. Dai, and R. W. Heath, Jr., "Successive bayesian reconstructor for channel estimation in fluid antenna systems," *IEEE Trans. Wireless Commun.*, 2025.

[5] C. Skouroumounis and I. Krikidis, "Fluid antenna with linear MMSE channel estimation for large-scale cellular networks," *IEEE Trans. Commun.*, 2023.

[6] T. Parr, G. Pezzulo, and K. J. Friston, *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. MIT Press, 2022.